

No to text conversion:

102 ➔ One Zero Two

```
#include<stdio.h>
```

```
#include<conio.h>
```

```
void main()
```

```
{
```

```
long n,rev=0,m; int r;
```

```
clrscr();
```

```
printf("Enter n value "); scanf("%ld",&n);
```

```
if(n<0)printf("-",n=-n); m=n;
```

```
while(n!=0){r=n%10; rev=rev*10+r; n=n/10;} /* rev no */
```

```
do
```

```
{
```

```
switch(rev%10)
```

```
{
```

```
case 0: printf("Zero");break;
```

```
case 1: printf("One");break;
```

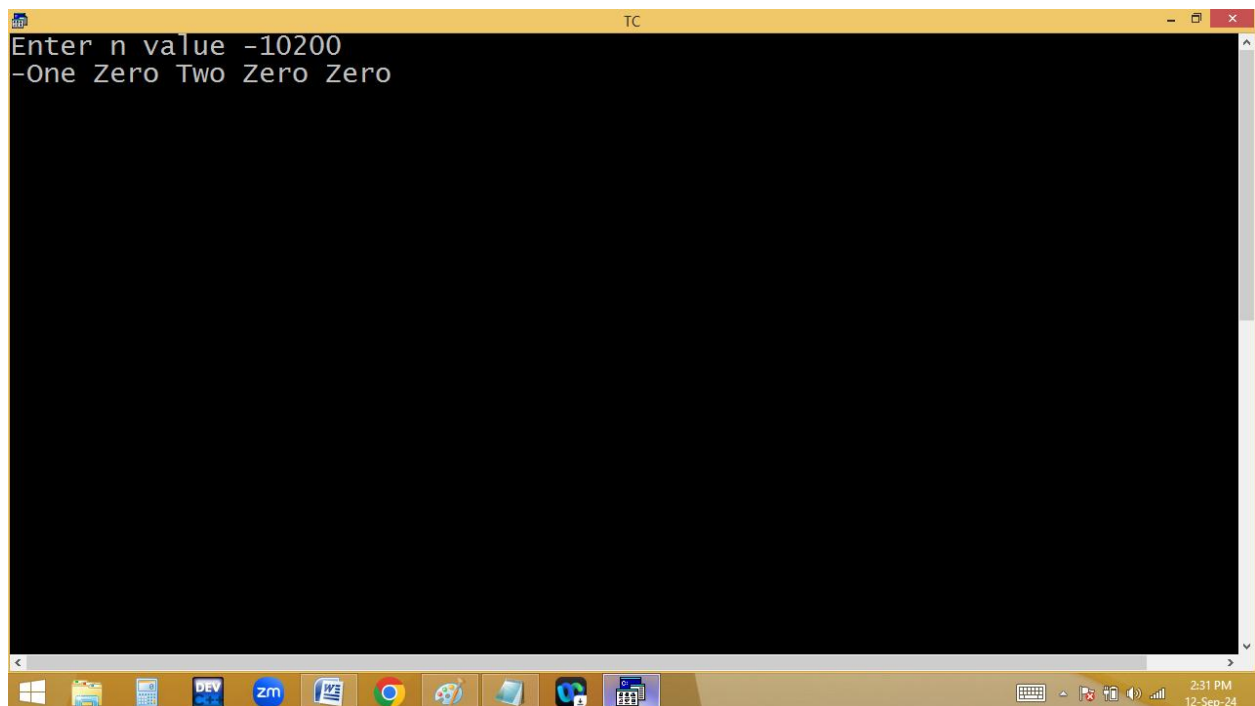
```
case 2: printf("Two");break;
```

```
case 3: printf("Three");break;
```

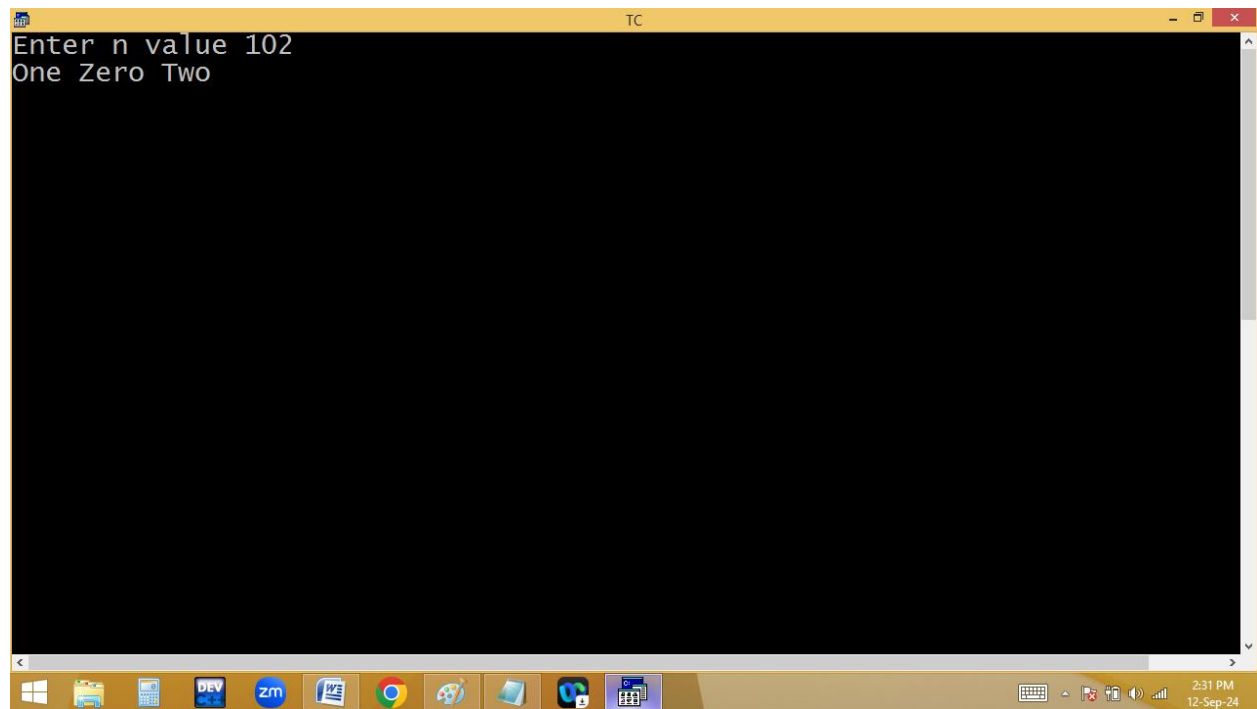
```
case 4: printf("Four");break;
```

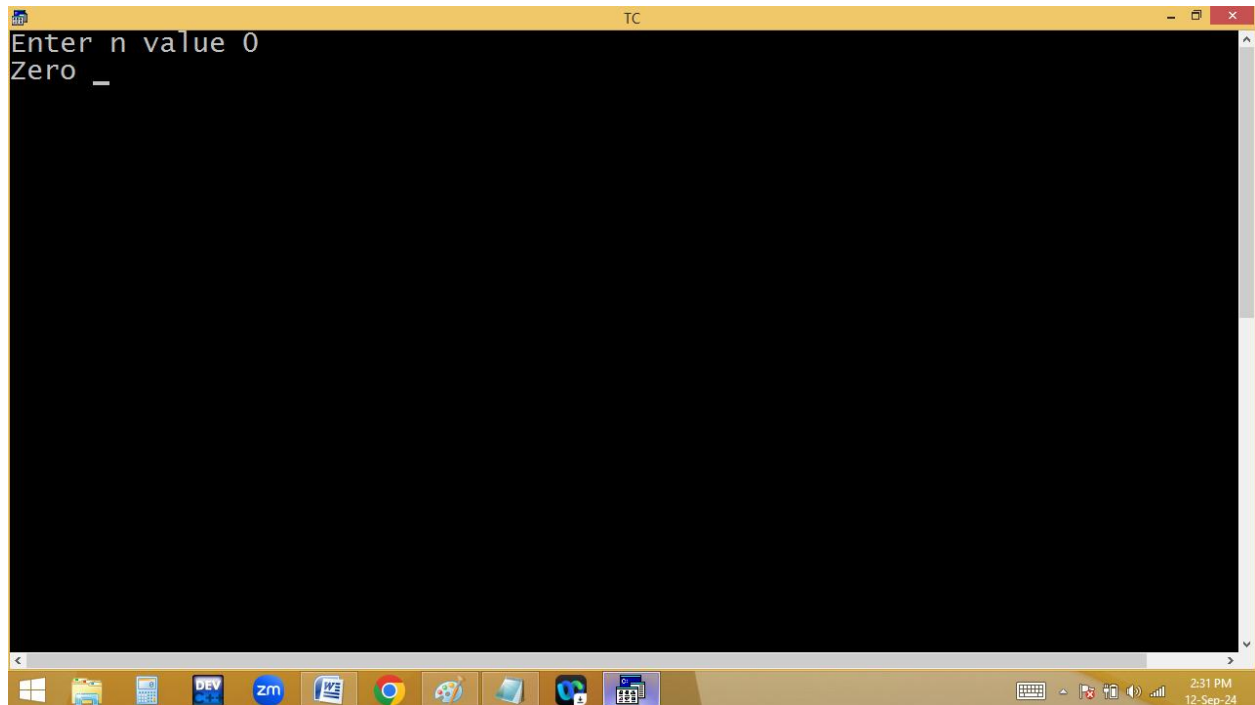
```
case 5: printf("Five");break;
case 6: printf("Six");break;
case 7: printf("Seven");break;
case 8: printf("Eight");break;
case 9: printf("Nine");
}

printf(" "); rev=rev/10;
}while(rev!=0);
while(m%10==0&& m!=0)printf("Zero ",m=m/10);
getch();
}
```



```
TC
Enter n value -10200
-One Zero Two Zero Zero
```





```
do
{
switch(rev%10)
{
case 0: p(zero);b;
case 1: p(one);b;
case 2: p(two);b;
case 9: p(nine);
}
rev=rev/10; p(" ");
}while(rev!=0);
```

n
102

rev

201 % 10 = 1 One ✓
201 % 10 = 0 Zero ✓
~~2~~ % 10 = 2 Two ✓
0

n = m
0

m

100 % 10 = 0

One Zero Zero

while(m%10==0 && m!=0) p("Zero ", m=m/10);

Finding Armstrong no:

The image shows two screenshots of the Turbo C++ (TC) IDE. The top screenshot displays the source code for a program that checks if a number is an Armstrong number. The code includes headers for stdio, conio, and math, and uses a loop to calculate the sum of the cubes of the digits. The bottom screenshot shows the program's execution, where the user has entered the value 1, and the program has outputted "Armstrong no".

```
File Edit Run Compile Project Options Debug Break/watch
Line 14 Col 2 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
#include<math.h>
void main()
{
int n,m,r,c=0,s=0;
clrscr();
printf("Enter n value "); scanf("%d",&n); m=n;
while(m!=0){c++; m=m/10;} /*counting no of digits */
m=n;
while(m!=0){r=m%10;s+=pow(r,c);m=m/10;}
puts(n==s?"Armstrong no":"Not an Armstrong no");
getch();
}
```

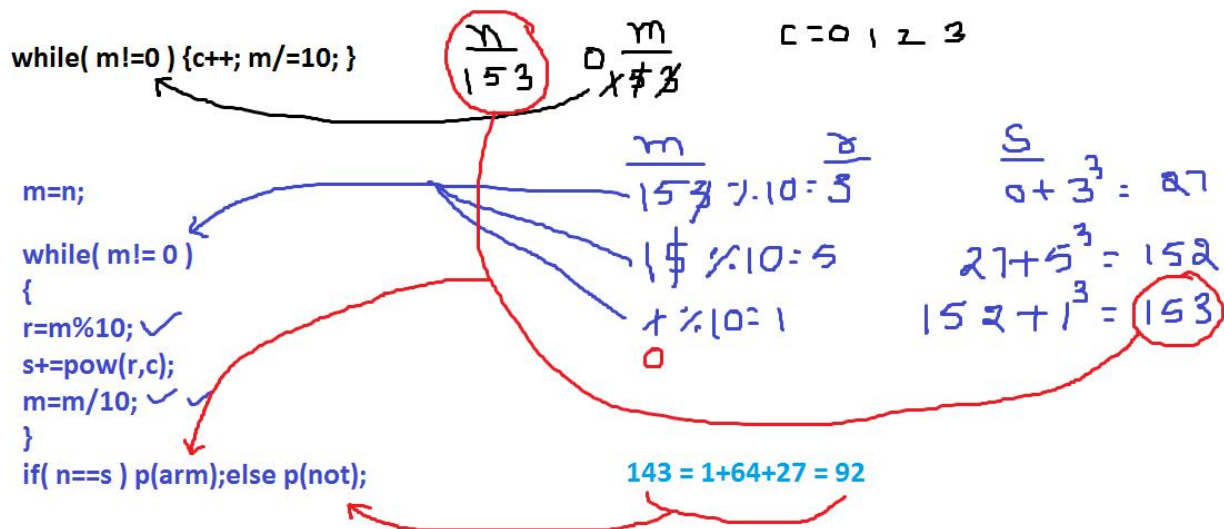
Enter n value 1
Armstrong no

```
TC
Enter n value 9
Armstrong no
```

```
TC
Enter n value 153
Armstrong no
```

```
TC
Enter n value 8208
Armstrong no
_
```

```
TC
Enter n value 143
Not an Armstrong no
_
```



for loop:

It is an entry control loop.

for is a keyword.

It is also used to repeat a program several times based on a condition.

When compared with while and do while, for loop is looking to be smart. In for it is compulsory to maintain two semicolons.

For works without condition also and default condition is always 1 i.e. true.

Generally for loop is having 3 expressions.

- 1. Initialization**
- 2. Test condition / expression**
- 3. Increment/decrement / updation**

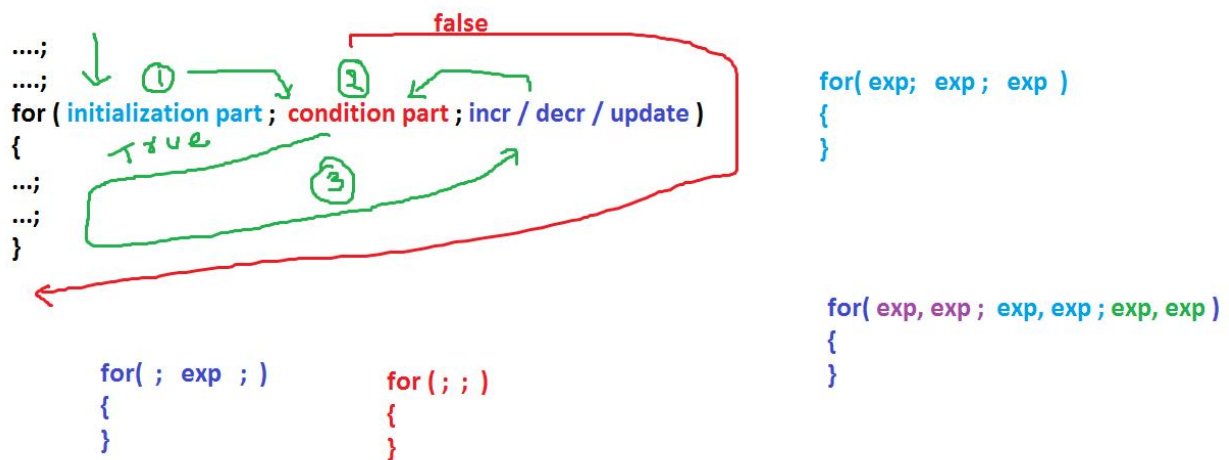
At first entry of for loop the initialization part is executed and later the test condition is checked. If the condition is true then the for block statements are executed. After completion of the block, the increment or decrement part is executed. Later once again the test condition is evaluated. If it is true then once again for block statements are executed. Like this the process is continued until the condition becomes false. Here the

initialization part is executed only once, at the time of loop beginning.

It is mandatory to maintain 2 semicolon (;) in a for loop.

If the for loop is having more than three expressions, it is mandatory to separate the expressions with , separator.

If the for loop is having less than three expressions, then leave the expressions with empty semicolon.



Printing given table:

For example 9th table

9*1=9

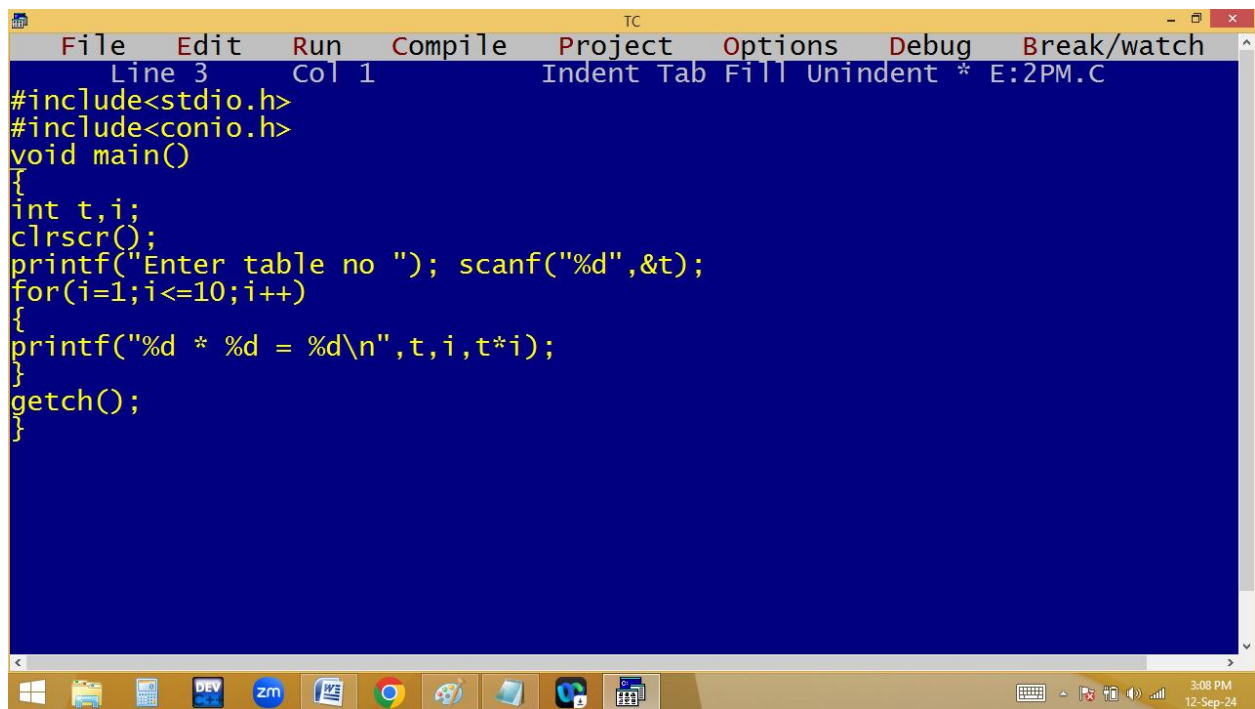
9*2=18

9*3=27

..

..

9*10=90



The screenshot shows a Turbo C++ (TC) IDE window. The menu bar includes File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. The status bar at the top indicates 'Line 3 Col 1' and 'Indent Tab Fill Unindent * E:2PM.C'. The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int t,i;
    clrscr();
    printf("Enter table no "); scanf("%d",&t);
    for(i=1;i<=10;i++)
    {
        printf("%d * %d = %d\n",t,i,t*i);
    }
    getch();
}
```

The Windows taskbar at the bottom shows various icons including the Start button, File Explorer, Calculator, DEV, ZM, Word, Chrome, Paint, and others. The system clock in the bottom right corner displays '3:08 PM' and '12-Sep-24'.

```
TC
Enter table no 9
9 * 1 = 9
9 * 2 = 18
9 * 3 = 27
9 * 4 = 36
9 * 5 = 45
9 * 6 = 54
9 * 7 = 63
9 * 8 = 72
9 * 9 = 81
9 * 10 = 90
```

```
TC
Enter table no 100
100 * 1 = 100
100 * 2 = 200
100 * 3 = 300
100 * 4 = 400
100 * 5 = 500
100 * 6 = 600
100 * 7 = 700
100 * 8 = 800
100 * 9 = 900
100 * 10 = 1000
```

$$\frac{t}{9} \times \frac{i}{1} = \frac{t \times i}{9}$$

```

for( i=1; i<=10; i++ )
{
    9 * 1 = 9
    p(" %d * %d = %d\n", t, i, t * i);
}

```

Findig perfect no or not? Sum of factors is equal to given no

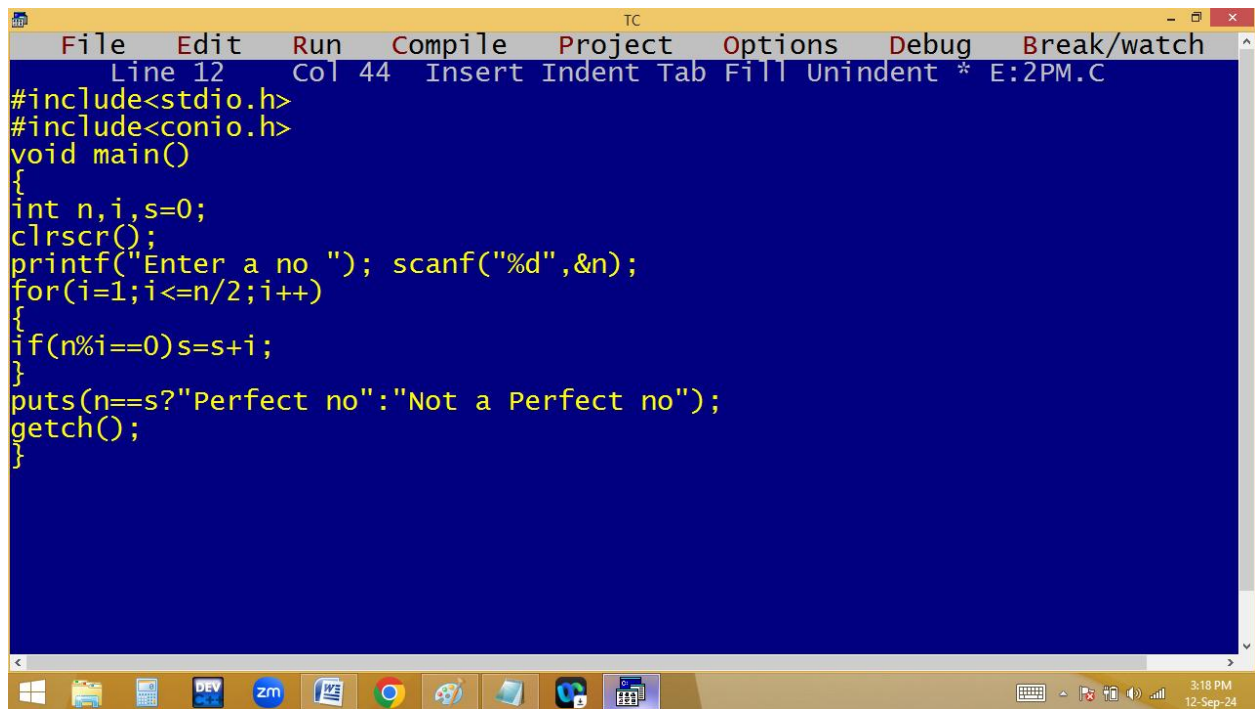
6 factors sum is $1+2+3 = 6$

28 factors sum is $1+2+4+7+14=28$

$$\begin{aligned}
 6 \% 1 &= 0 \\
 6 \% 2 &= 0 \\
 6 \% 3 &= 0
 \end{aligned}$$

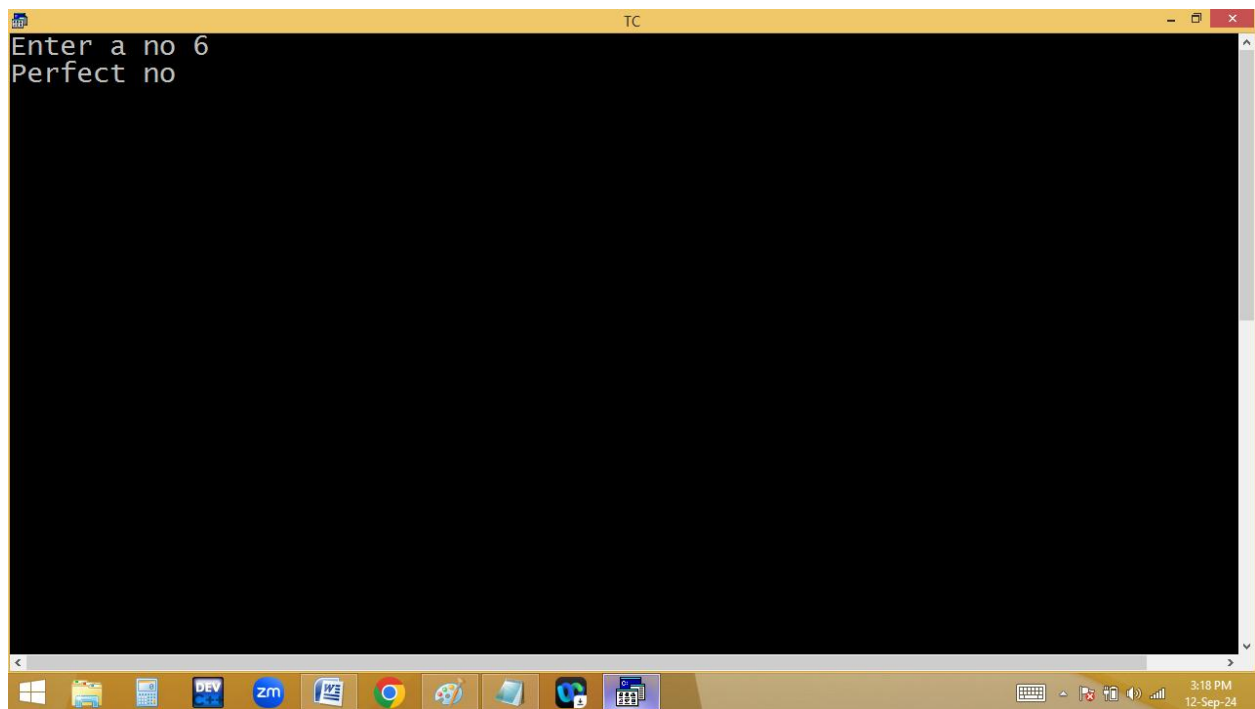
6

4 factors sum is $1+2=3$ $\leftarrow$ not a perfect no



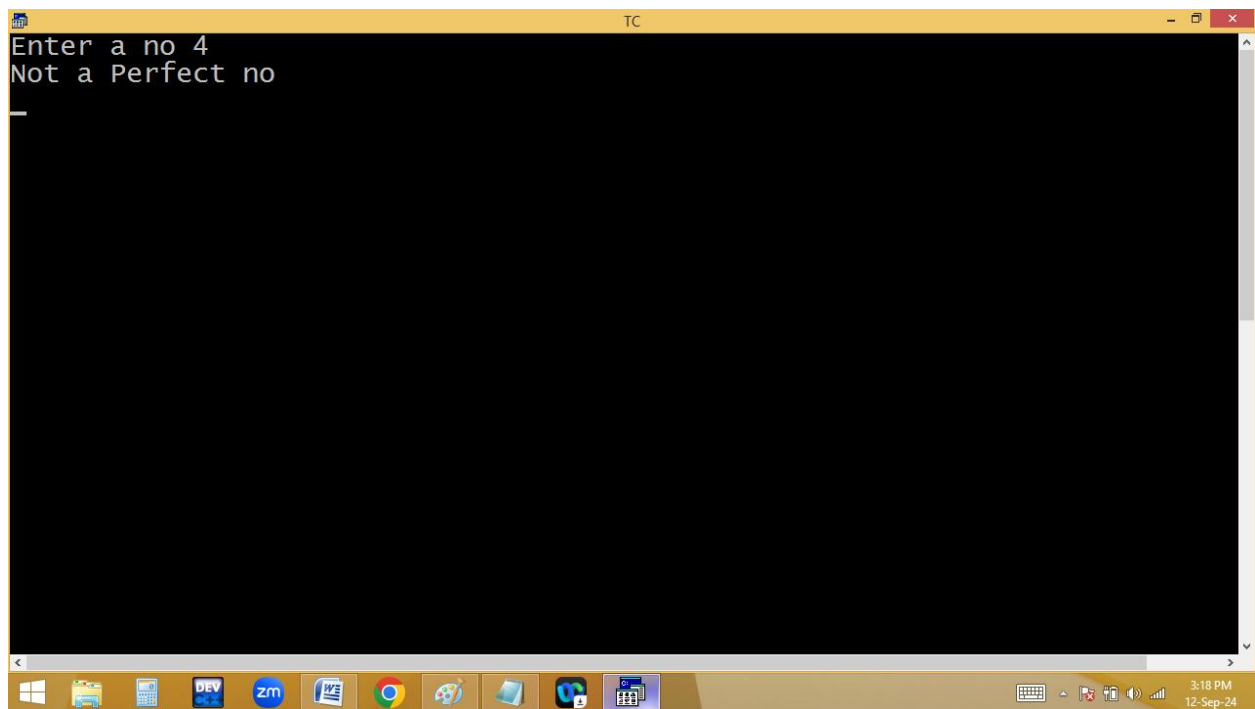
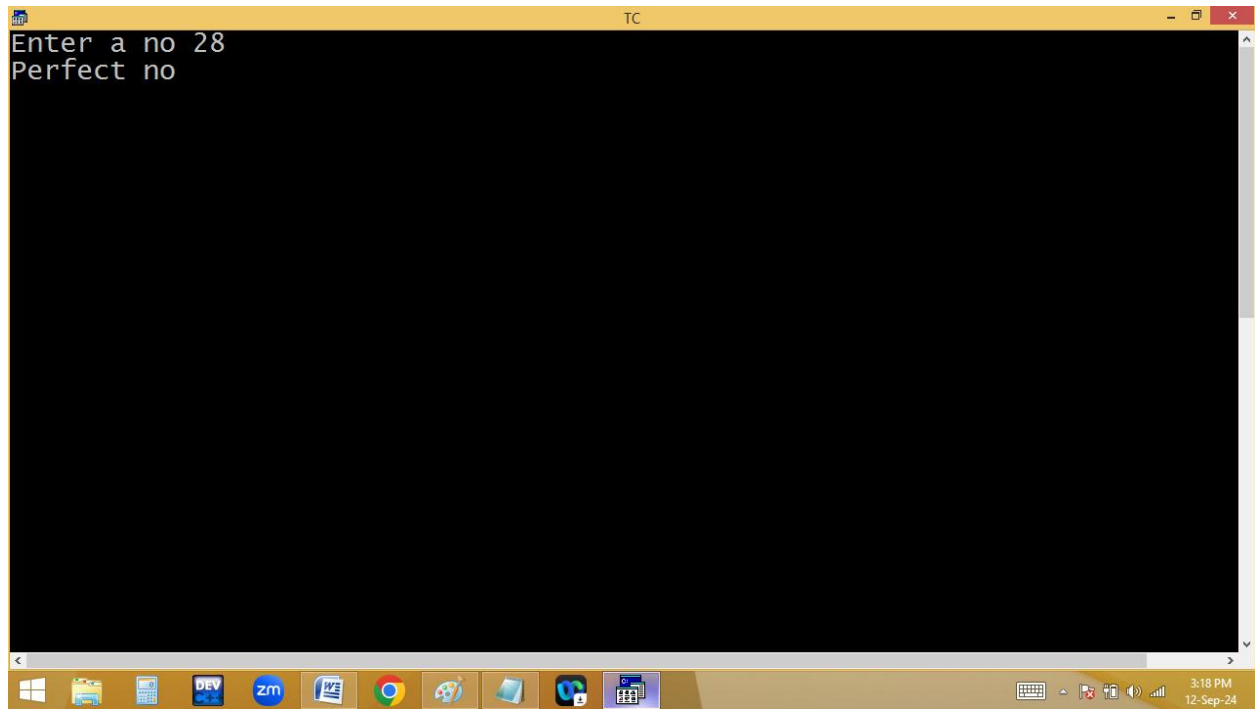
The screenshot shows the Turbo C++ (TC) IDE with a blue background. The menu bar includes File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. The status bar at the bottom indicates Line 12, Col 44, and the file name E:2PM.C. The code is as follows:

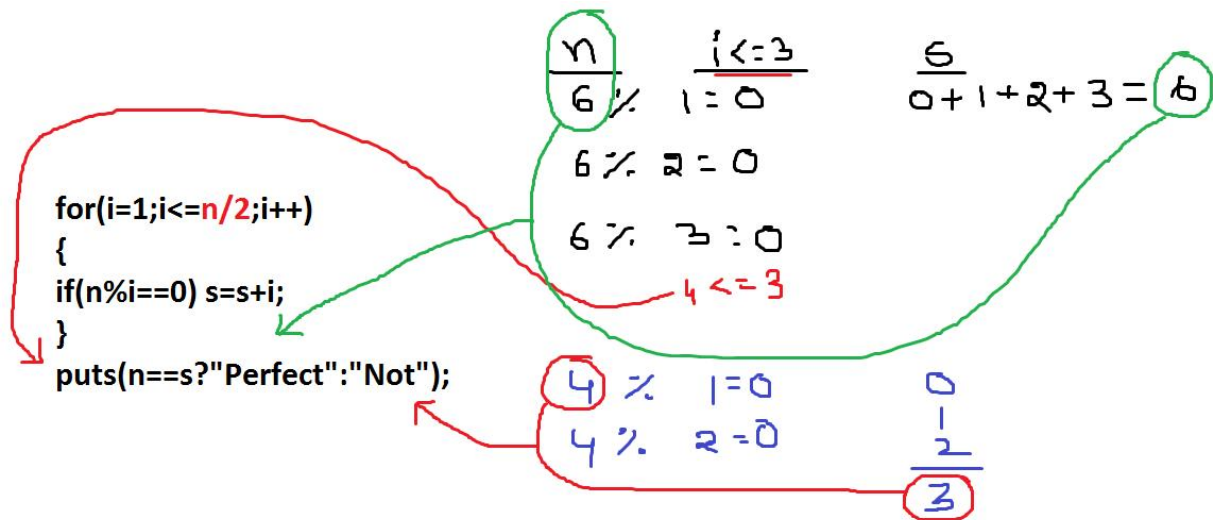
```
Line 12 Col 44 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int n,i,s=0;
clrscr();
printf("Enter a no "); scanf("%d",&n);
for(i=1;i<=n/2;i++)
{
if(n%i==0)s=s+i;
}
puts(n==s?"Perfect no":"Not a Perfect no");
getch();
}
```



The screenshot shows the Turbo C++ (TC) IDE with a black background. The program has executed, and the output is displayed in the top-left corner of the window:

```
Enter a no 6
Perfect no
```





Finding prime / composite no?

count of factors is 2 or the no divisible with 1 and itself only is called prime.

$$2 \% 1 = 0$$

$$2 \% 2 = 0$$

$$3 \% 1 = 0$$

$$3 \% 3 = 0$$

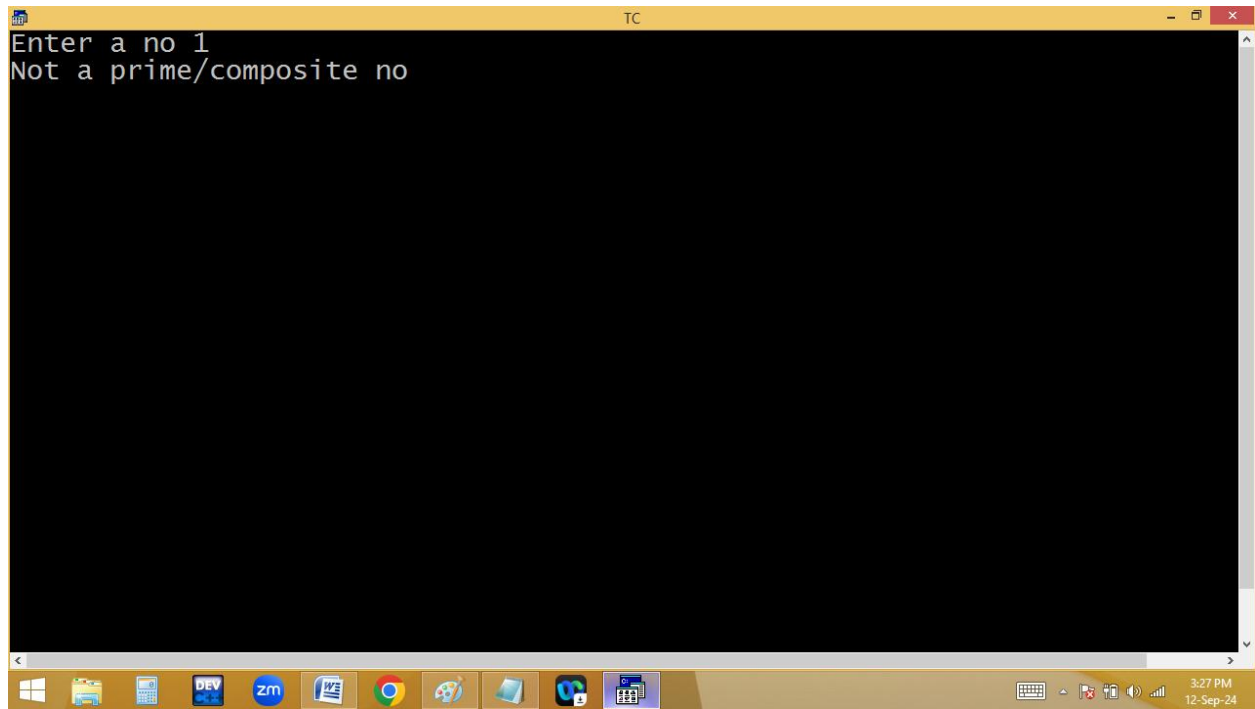
$$4 \% 1 = 0$$

$$4 \% 2 = 0$$

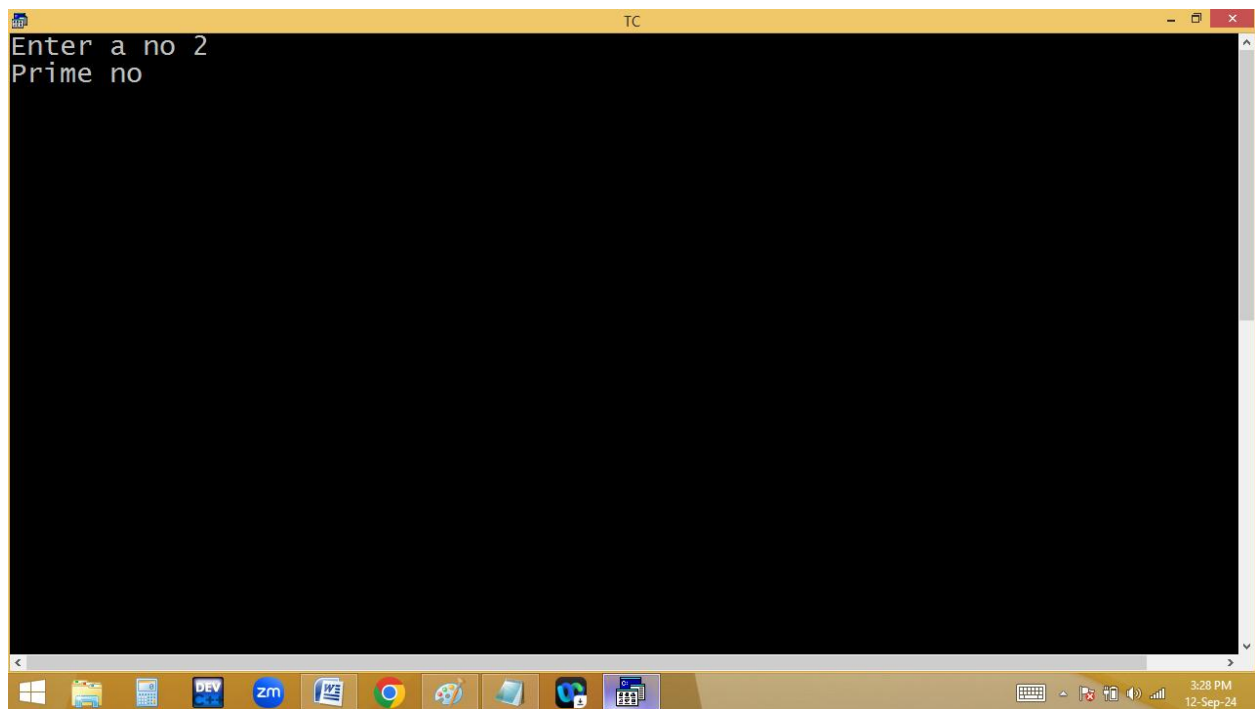
$$4 \% 4 = 0$$

```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 16 Col 2 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int n,i,c=0;
clrscr();
printf("Enter a no "); scanf("%d",&n);
if(n==1)puts("Not a prime/composite no");
else
{
for(i=1;i<=n;i++)
{
if(n%i==0)c++;
}
puts(c==2?"Prime no":"Composite no");
}
getch();
}
```

```
TC
Enter a no 1
Not a prime/composite no
```

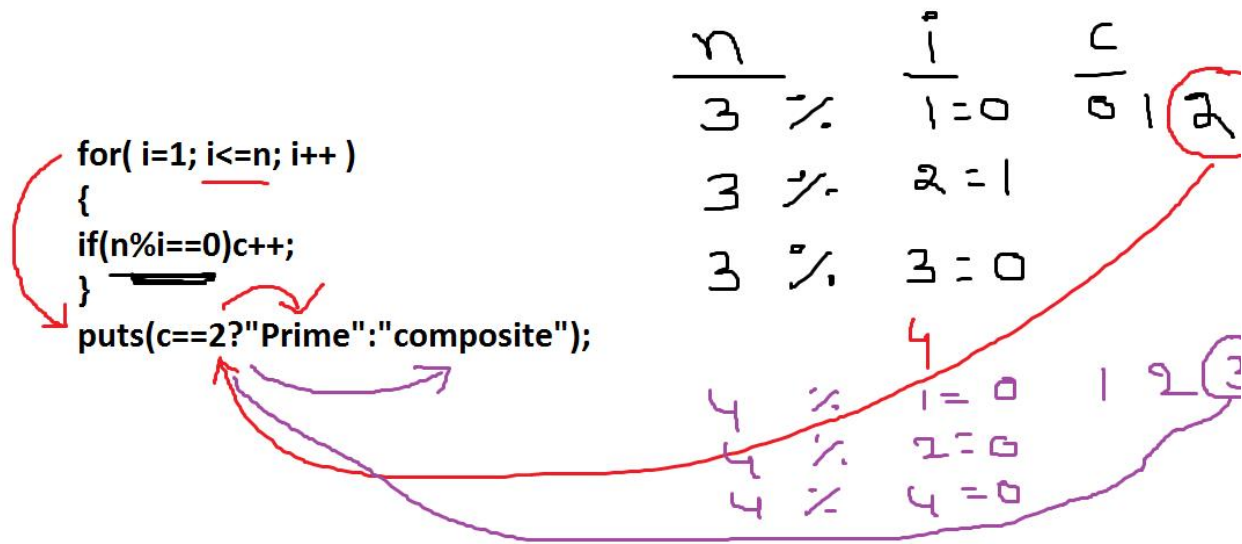


```
TC
Enter a no 2
Prime no
```



```
TC
Enter a no 3
Prime no
```

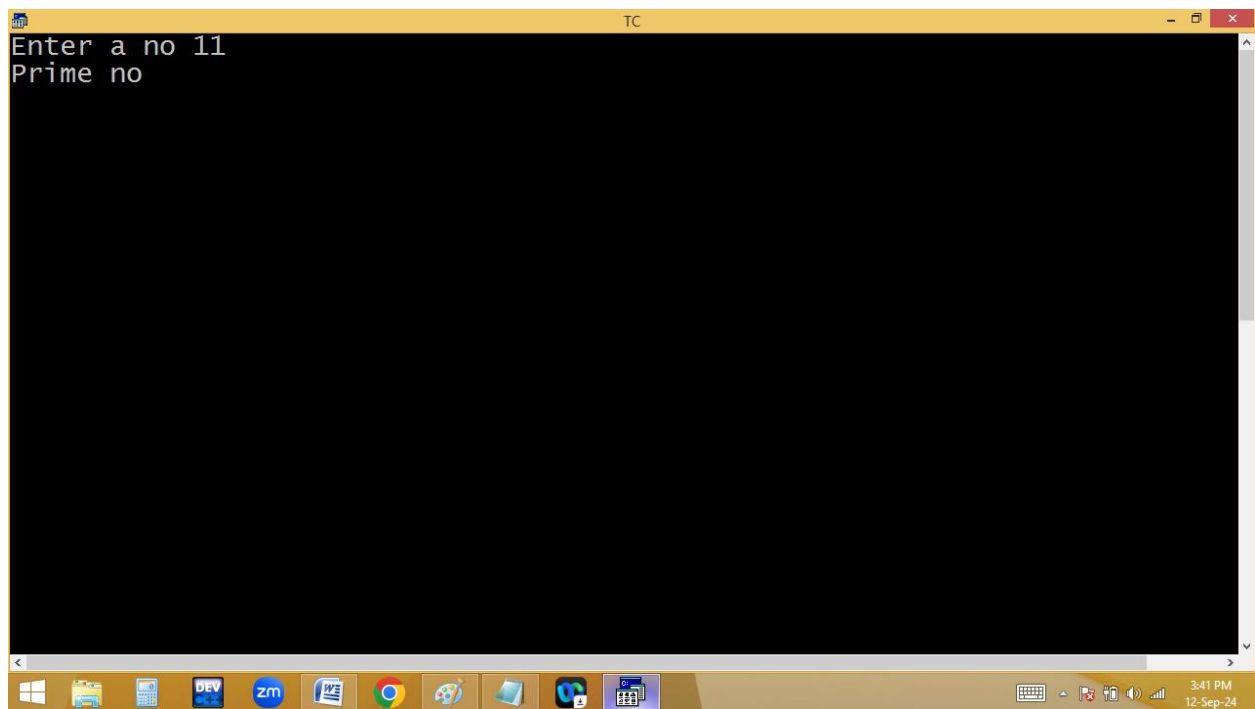
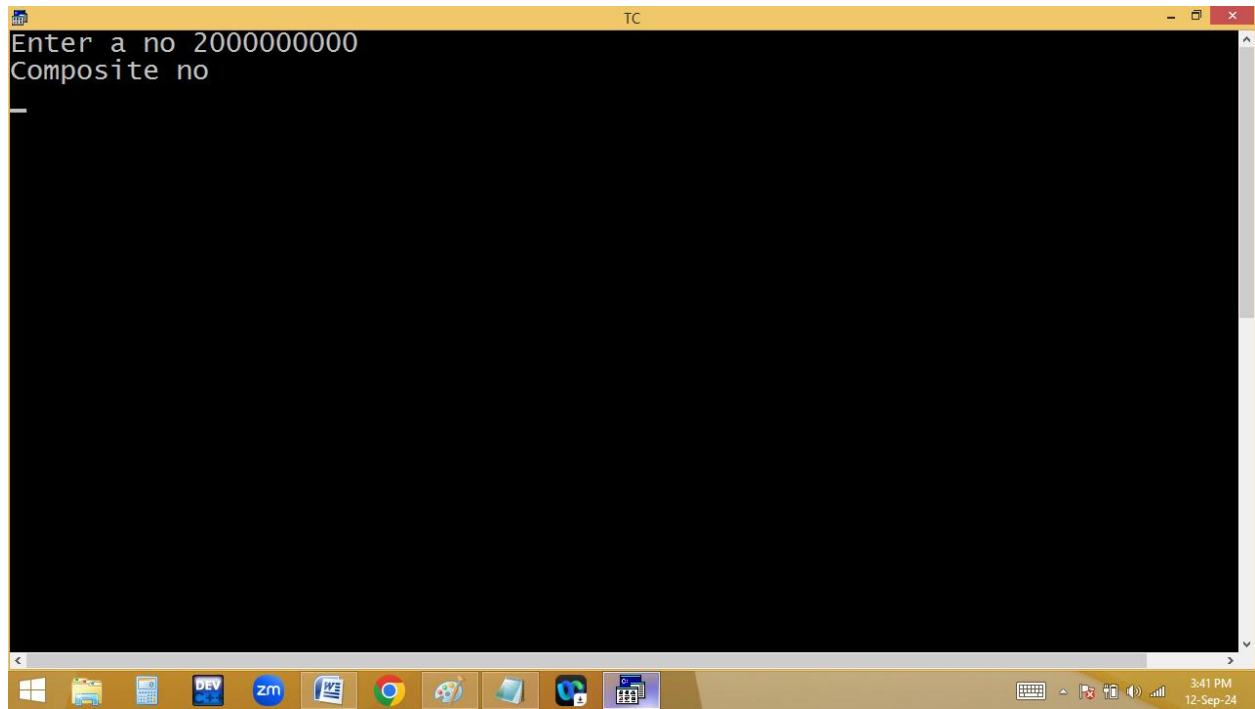
```
TC
Enter a no 4
Composite no
_
```

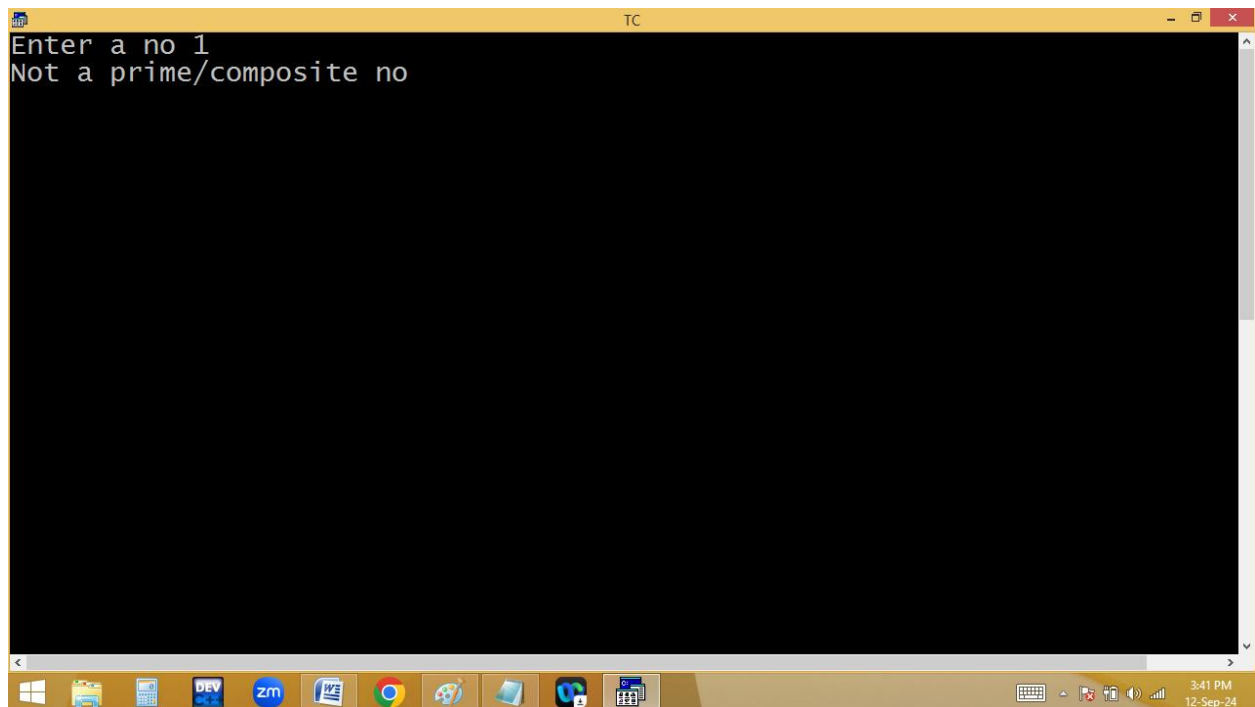


```

File Edit Run Compile Project Options Debug Break/watch
Line 15 Col 16 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
long int n,i;
clrscr();
printf("Enter a no "); scanf("%ld",&n);
if(n==1)puts("Not a prime/composite no");
else
{
for(i=2;i<=n/2;i++)
{
if(n%i==0){puts("Composite no");getch(); return; }
}
puts("Prime no");
}
getch();
}

```



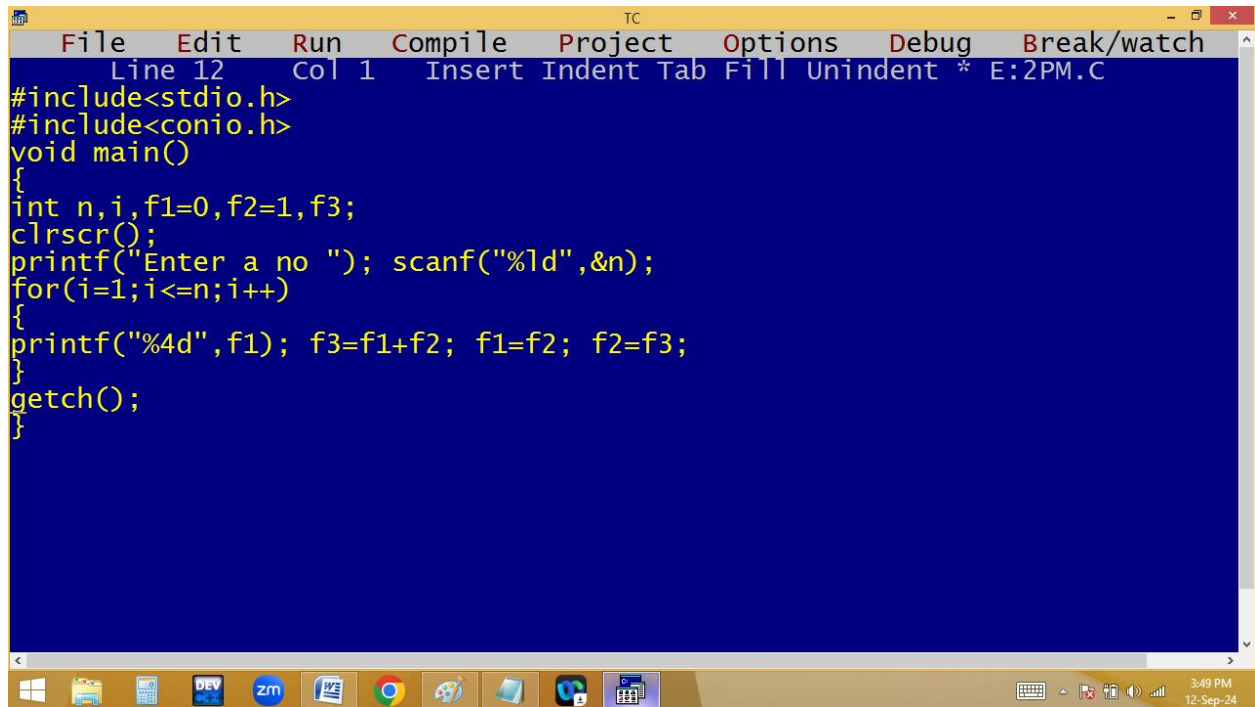


11 - ~~2~~ ~~3~~ ~~4~~ ~~5~~
10 - ~~1~~ 2 5 ~~10~~
100 - ~~1~~ 2 4 5 10 20 25 50 ~~100~~
5 - ~~1~~ ~~5~~

```
for( i=2; i<=n/2;i++)  
{  
    if(n%i==0)p("com"); return;  
}  
    p(prime);
```

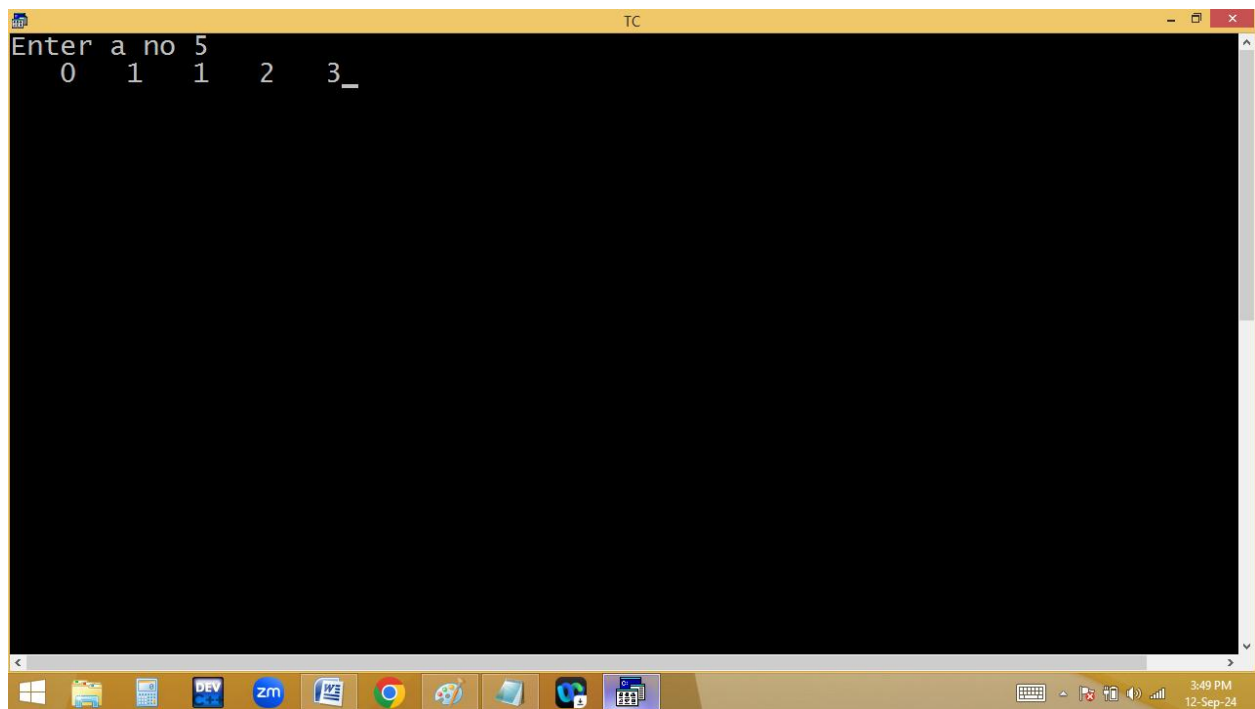
Fibonacci series:

N = 5 → 0 1 1 2 3



The screenshot shows the Turbo C++ (TC) IDE with a blue background. The menu bar includes File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. The status bar at the bottom indicates 'Line 12 Col 1' and 'Insert Indent Tab Fill Unindent * E:2PM.C'. The code is as follows:

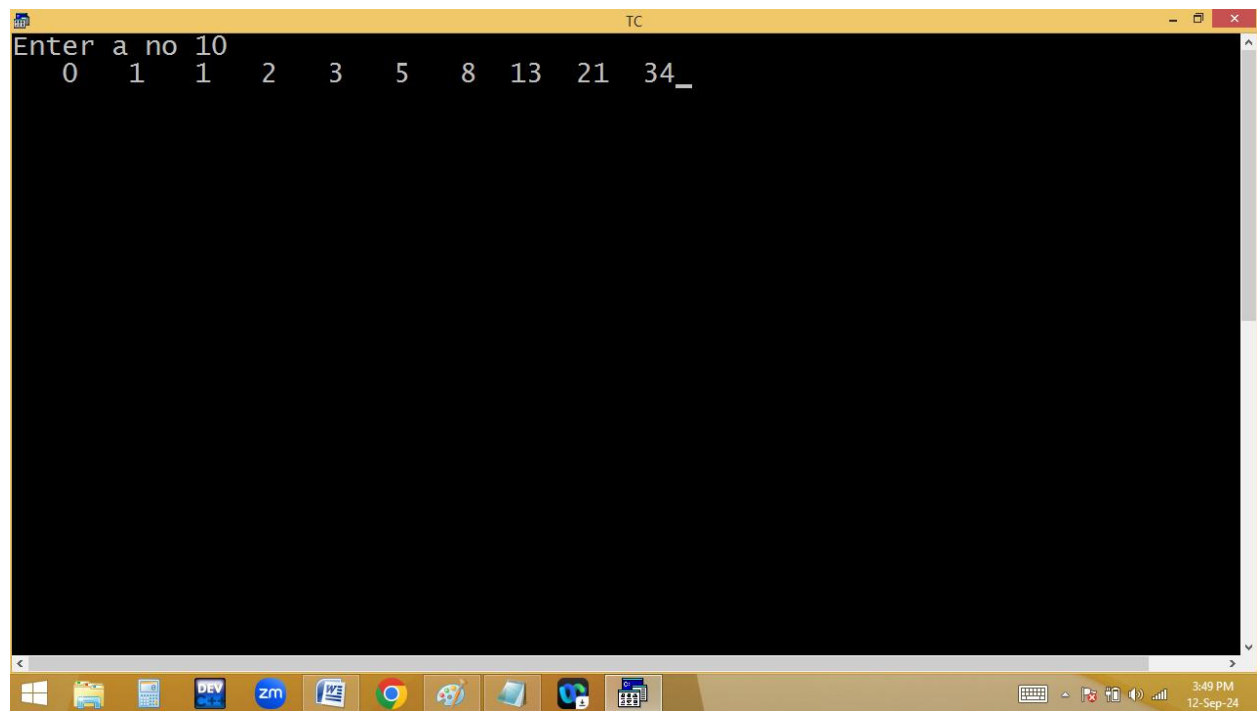
```
#include<stdio.h>
#include<conio.h>
void main()
{
int n,i,f1=0,f2=1,f3;
clrscr();
printf("Enter a no "); scanf("%ld",&n);
for(i=1;i<=n;i++)
{
printf("%4d",f1); f3=f1+f2; f1=f2; f2=f3;
}
getch();
}
```



The screenshot shows the Turbo C++ (TC) IDE with a black background. The program has executed, and the output is displayed as follows:

```
Enter a no 5
0    1    1    2    3_
```

```
TC
Enter a no 10
0 1 1 2 3 5 8 13 21 34_
```



The screenshot shows a Windows desktop environment. A terminal window titled "TC" is open, displaying a program that prompts the user to "Enter a no". The user has entered "10". The program then outputs the Fibonacci sequence up to the 10th term: "0 1 1 2 3 5 8 13 21 34_". The taskbar at the bottom shows various application icons, including Windows Explorer, DEV, zm, and Chrome. The system clock in the bottom right corner indicates the time is 3:49 PM on 12-Sep-24.